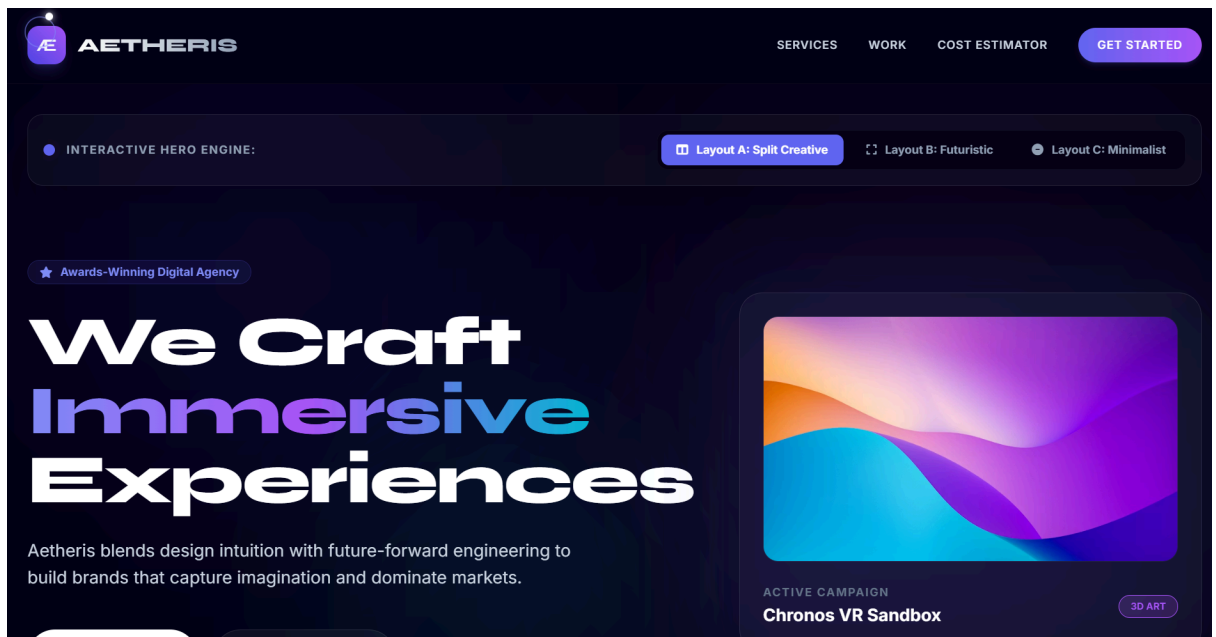


1. Weak Baseline Prompt

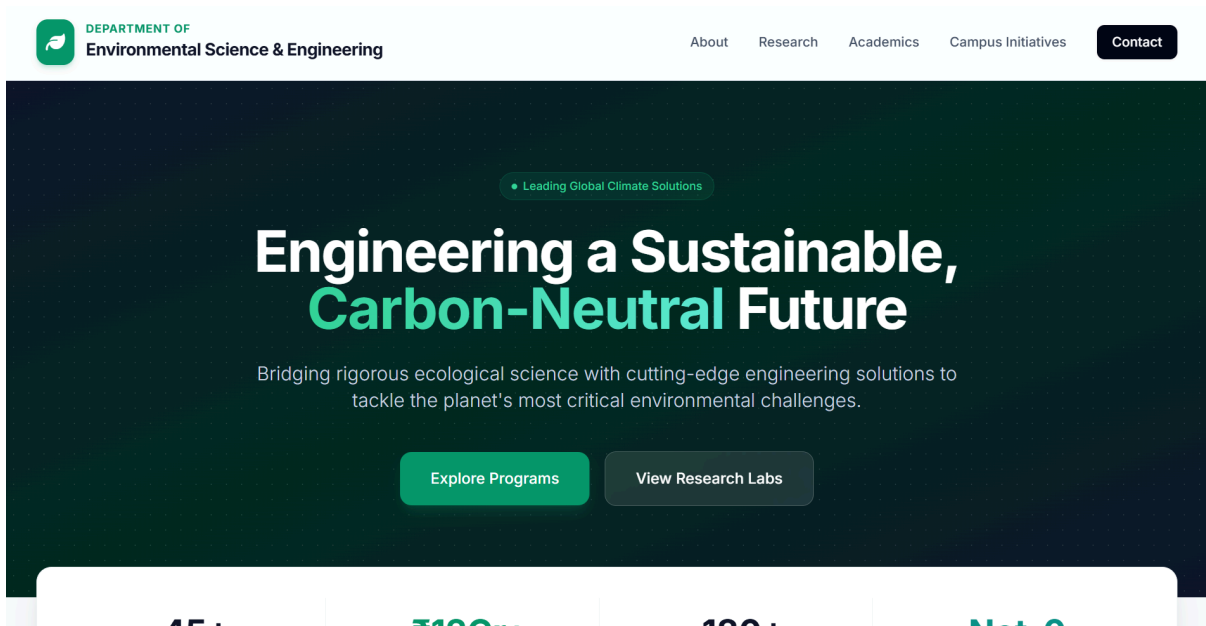
Prompt: "can u create a website for me"

- **Output Received:** A generic HTML template with standard placeholders, basic layout.



Version 1: Adding a Clearer Goal & Subject

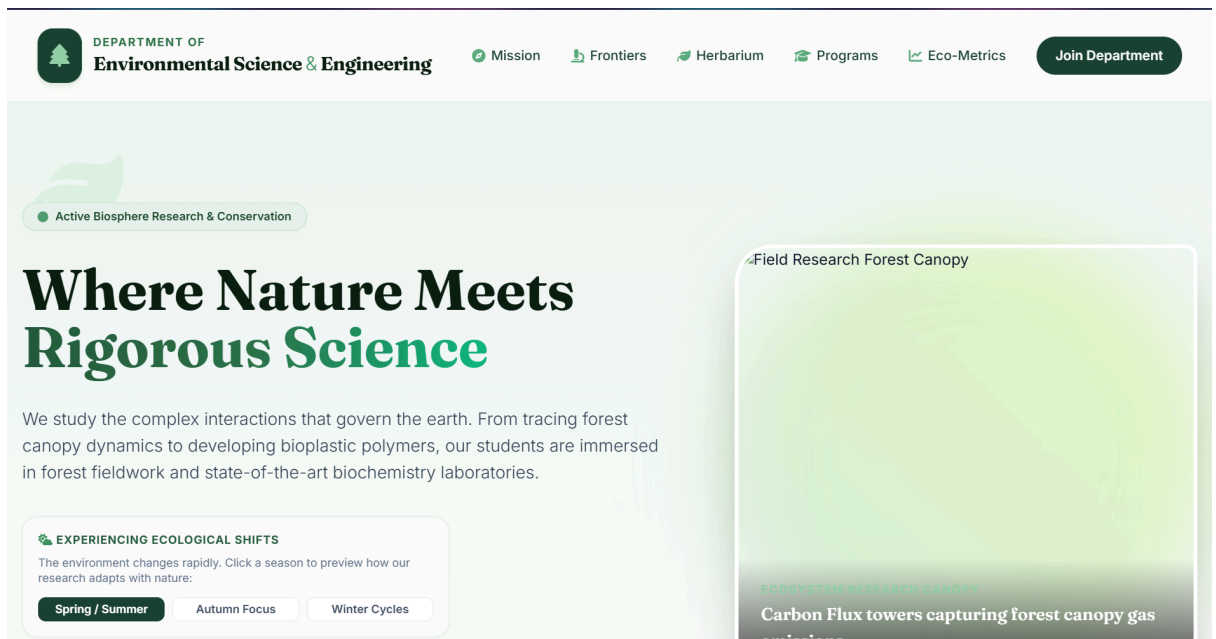
- **Prompt:**
"I need the website for my institute's department, Environmental Science and Engineering. "



1. **What changed in the prompt:** the website shifted from a completely open ended request to a specific academic domain (Environmental Science & Engineering department website).
2. **What actually improved in the output:** The AI generated a structured academic landing page with a Hero section, research metrics, a course list, and a basic sustainability baseline calculator.
3. **What still failed:** The visual identity was bad, like boring. Utilizing flat gray and black color instead of reflecting natural themes.
4. **What to try next:** Add sensory and stylistic constraints focused on nature, botanical colors, and organic shapes.

Version 2: Adding Aesthetic Constraints (Nature-Theme & Colors)

- **Prompt:**
"the website should have environmental, nature, and sustainability elements so that its easier to know that its the department website. Add some shades of leaf green around the website. For now it doesnt look like that."

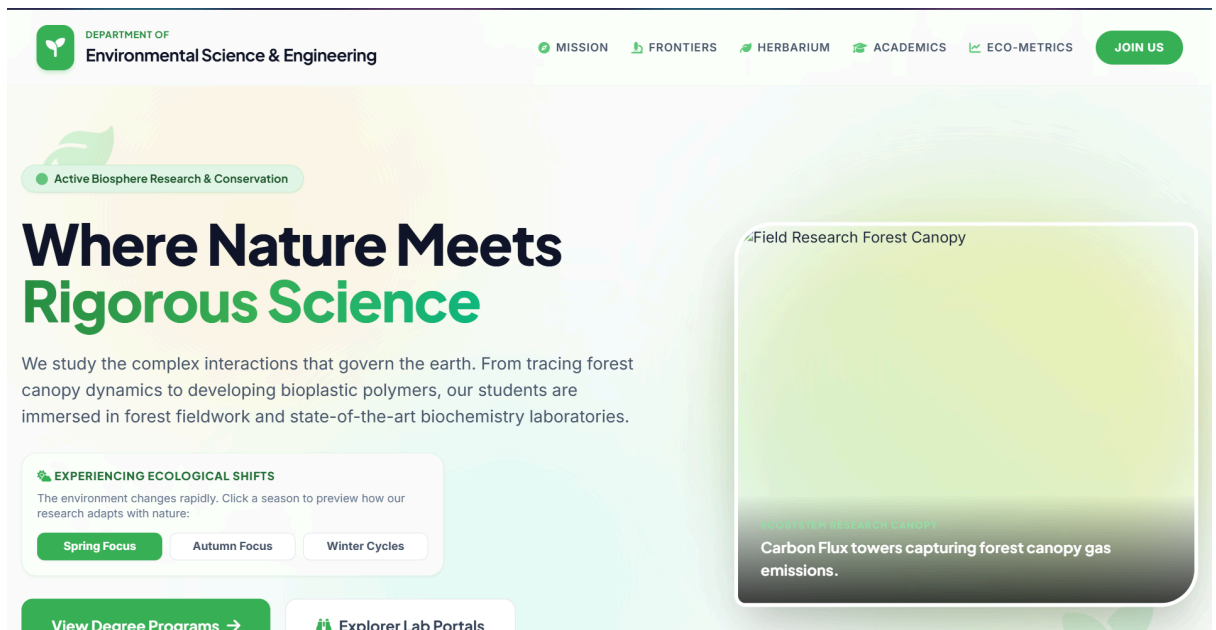


1. **What changed in the prompt:** added aesthetics (nature-themed elements, custom leaf green palette, and specific sustainability widgets).
2. **What actually improved in the output:** the site adopted an organic green theme, and the AI added highly thematic interactive elements like a "Digital Herbarium" and real "Field Station" data monitors.
3. **What still failed:** The green was way too dark/murky, typography felt generic like normal in other websites, there were no visual animations, and the page was like really bad due to browser scroll dynamics.
4. **What to try next:** Refine the color spectrum, specify concrete font pairings, and introduce a modern smooth scrolling library with floating background animations.

Version 3: Adding Quality Criteria & Technical Libraries (Locomotive & Typography)

- **Prompt:**

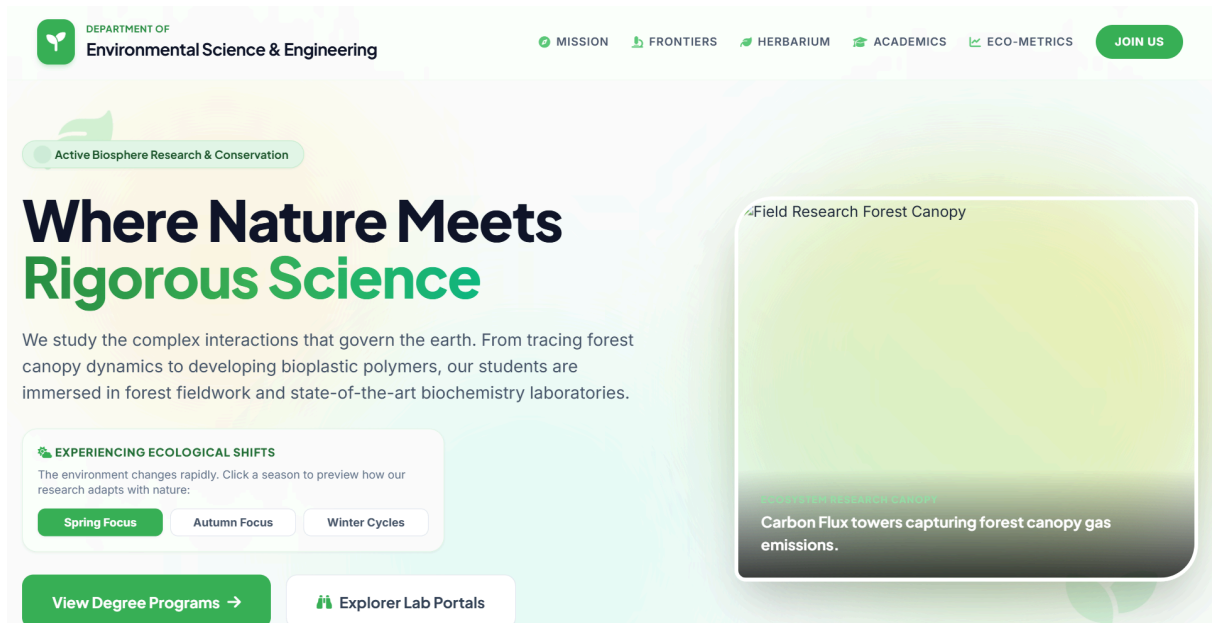
"the color isnt just doing right for me. The green you have used is way too dark. I want you to make it lighter and a more bright green, but not too bright. Also, tweak the font to Headings: Plus Jakarta Sans Body: Inter. Also i need a few animations going around in the background with smooth scrolling enabled. You can use locomotive scroll for this."



1. **What changed in the prompt:** specified concrete font pairings (Plus Jakarta Sans for Headings, Inter for Body), demanded a precise color balance (brighter sprout-green), and integrated a complex third-party library (Locomotive Scroll) for smooth scrolling.
2. **What actually improved in the output:** the scrolling physics became professional. Apply background glow blobs floated softly in the background to simulate light filtering through a forest canopy.
3. **What still failed:** The layout felt visually empty and plain because secure images were missing (browser previews were actually blocking <http://> placeholder assets).
4. **What to try next:** Feed strict requirements for high-resolution, secure (<https://>) environmental images and an integrated photography layout.

Version 4: Adding Media & Content Constraints (Secure Gallery)

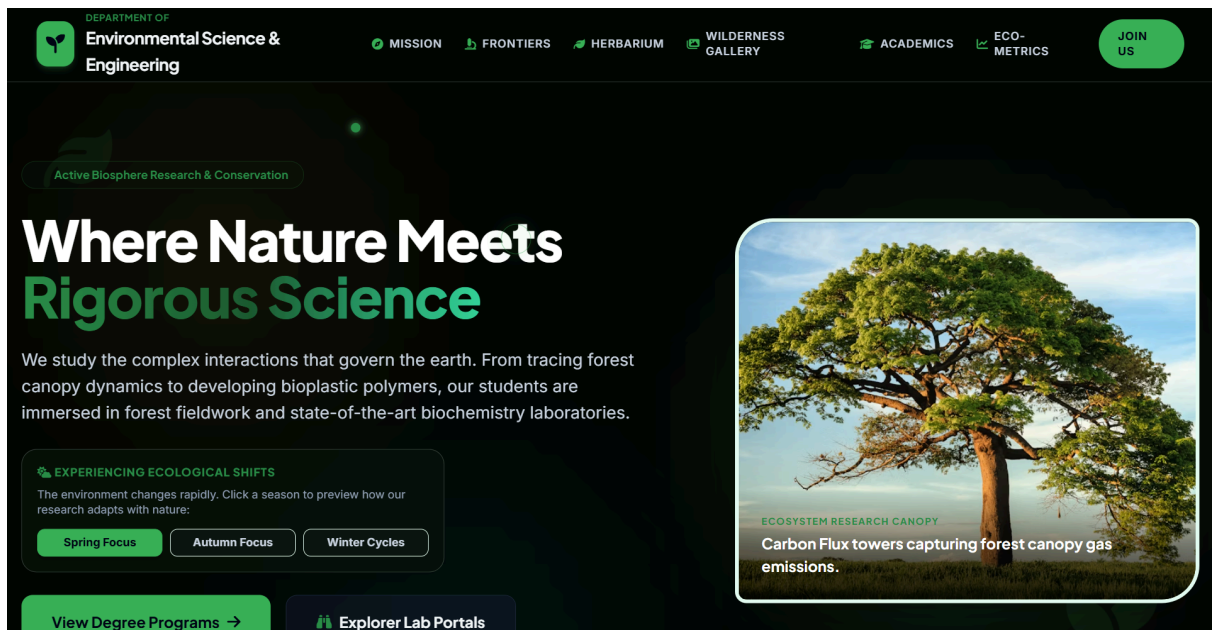
- **Prompt:**
"the website looks great but where are the photos? i cant see any photos. I need more photos on the website or it looks kinda plain, ngl."



1. **What changed in the prompt:** explicitly requested the addition of photos, but few photos still didn't show up.
2. **What actually improved in the output:** replaced all broken placeholders with high-quality, secure (<https://>) Unsplash forest canopy, fieldwork, and lab photos, and built a dedicated "Wilderness Gallery" grid to showcase active on-site science.
3. **What still failed:** The bright design was highly glaring and strained the eyes during extended testing. Furthermore, the default system cursor felt disconnected from the premium kinetic scrolling experience.
4. **What to try next:** transition to a dark mode design palette and engineer a custom, smooth-tracking cursor follower.

Version 5: Adding UX Variations & Custom Interactive States (Dark Theme & LERP Cursor)

- **Prompt:**
"lovely. Now convert this to dark theme because there is too much strain on the eyes. Also, add a smooth cursor follower with a green color, it can be like a blob of paint or a circle whatever looks minimalistic."



1. **What changed in the prompt:** Requested a dark mode transformation ("dark bio-theme") and specified a minimalistic, custom mouse pointer follower, prompt was generic but output was better than I expected.
2. **What actually improved in the output:** The site converted to a premium dark theme with glassmorphic cards and bright glowing highlights. A custom mouse-ring follower with high-frequency linear interpolation (LERP) physics was successfully engineered.
3. **What still failed:** Clicking on the interactive controls (changing plants or weather seasons) triggered abrupt, instant display swaps which felt disjointed compared to the butter smooth scrolling. Also, call to action buttons had broken links, like Join Us should take us to the official website.

The Final Prompt that anyone can use

Create a single-file, production-grade landing page for a university "Department of Environmental Science & Engineering" using HTML, Tailwind CSS, FontAwesome icons, Google Fonts, and Locomotive Scroll.

The website must conform to the following specifications:

1. **THEME & PALETTE (Dark Bio-Theme):**
 - Background: Very deep organic green-black (#030704 / #09120c).
 - Primary Accents: Vibrant, fresh sprout/leaf green (#39b057 / #69e285) used for highlights, icons, borders, and glowing accents.

- Card styling: Frosted dark glassmorphism (glass cards with semi-transparent borders).
- Ambient Depth: Include absolute background blobs that drift and scale slowly in the background (using CSS animations) to simulate dappled light filtering through a canopy.

2. TYPOGRAPHY:

- Headings: Plus Jakarta Sans (bold, modern, tracking-tight).
- Body: Inter (highly legible, professional, clean sans-serif).

3. SCROLLING & UX PHYSICS (Locomotive Scroll & Cursor):

- Embed and initialize Locomotive Scroll for butter-smooth inertial scrolling. Implement parallax scrolling triggers on major section imagery.
- Include a dual-element custom kinetic cursor follower: a high-responsiveness center dot paired with a lazy, organic trailing ring using Linear Interpolation (LERP) inside a requestAnimationFrame loop.
- The cursor follower must gracefully scale up, glow, and change colors on hovering over links, buttons, and input elements. Hide custom cursors automatically on mobile devices.

4. SECTIONS & LIVE INTERACTIVITY:

- Sticky Header: Dynamic navigation with custom anchor scroll triggers matching Locomotive scroll physics. Include a "Join Us" button linked directly to "<https://www.esed.iitb.ac.in/>" which opens in a new tab.
- Hero Section: Striking introduction focusing on research and global climate solutions, featuring a "Season Theme Selector" that smoothly transitions the hero image (using scale and fade opacity delays) when clicking different seasons (Spring, Autumn, Winter).
- Research Areas: Grid showcasing core research domains (Atmospheric, Hydrobiology, Soil Networks) with expandability.
- Live Outpost Monitor: Interactive toggle buttons to preview different remote research outposts. Clicking a station must trigger a 250ms fade-out/scale-down animation, swap data stats (pH, moisture, temperature) and image, and smoothly fade/scale back in.
- Interactive Herbarium: A clickable registry cataloging local campus plants. Clicking tags must transition-fade the card contents to display unique botanical metadata, carbon indices, and photos.
- Campus Wilderness Gallery: A visual 4-column photo grid highlighting real-world field activities using secure, high-resolution Unsplash images.
- Sustainability Estimator: A functional carbon calculator. Use an input slider/field where sliding or inputting a custom tree quantity immediately calculates and displays annually sequestered CO2 kilograms with a gentle scale-up bounce animation on update.

5. MEDIA:

- Every image source must use secure, high-resolution HTTPS links from Unsplash, framed using custom leaf-card border radii.

